Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/04/2006
PI Name:	Urban, David L. Ph.D.		
Project Title:	Smoke Aerosol Measurement Experiment (SAME)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	216-433-2835
Organization Name:	NASA Glenn Research Center		
PI Address 1:	Mail Stop 77-5, Combustion and Reacting Systems Branch		
PI Address 2:			
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	03/02/2001	**End Date:**	02/28/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mulholland, George (University of Maryland) Yang, Jiann (National Institute for Standards and Technology) Cleary, Thomas (National Institute for Standards and Technology) Yuan, Zeng guang (National Center for Microgravity Research)		
Grant/Contract No.:	IH1013215		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project is a flight-definition investigation funded under the Bio-Astronautics initiative. The team is a combination of NASA Glenn Research personnel with experience in microgravity combustion and researchers from the National Institute for Standards and Technology and the University of Maryland who have extensive experience in smoke detection. The investigation will define an experiment that will make measurements of the particle size distribution of spacecraft smoke from a variety of spacecraft materials. The overall project objective is to improve the reliability of spacecraft fire detection by making measuremstns of the particle size distribution of spacecraft smoke. The smoke detectors currently used in NASA spacecraft are based upon understanding from 1-g smoke measurements. Experiments conducted aboard the space shuttle in 1996 demonstrated that the particle size distribution for various materials can change substantially and that smoke detector sensitivity in 1-g is not necessarily a good indicator for performance in low-gravity. This suggests that the reliability of spacecraft smoke detectors can be substantially improve dif they are designed based upon the smoke particulate properties of spacecraft smoke.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report received this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)